

Student Analytics Success

Business Problems & Data Mining Objectives

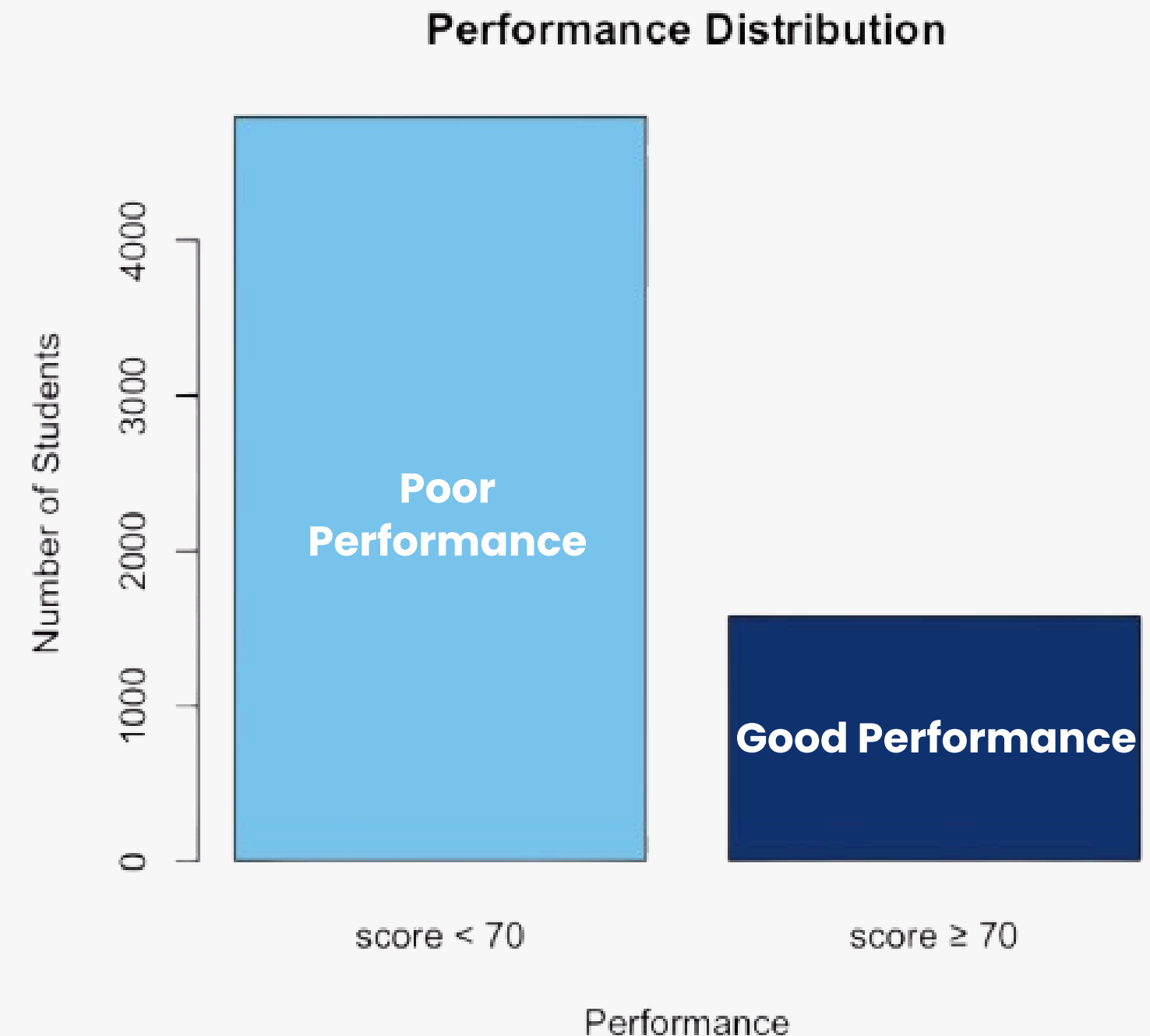
What we figured out ?

The business problem for educational institutions is that **underperformance** creates **long-term consequences** - lower results, higher drop-out rates, inefficient use of resources, and **poor academic reputation**.

Being able to predict which students are likely to perform poorly allows institutions to plan interventions, allocate support resources, and improve academic outcomes more effectively.

What we want to achieve ?

- Identify the correlated variables of student performance
- Evaluate and compare predictive models:
Logistic Regression, Decision Tree
- Develop targeted recommendations to **improve student outcomes** and **reduce academic underperformance**



Dataset Overview

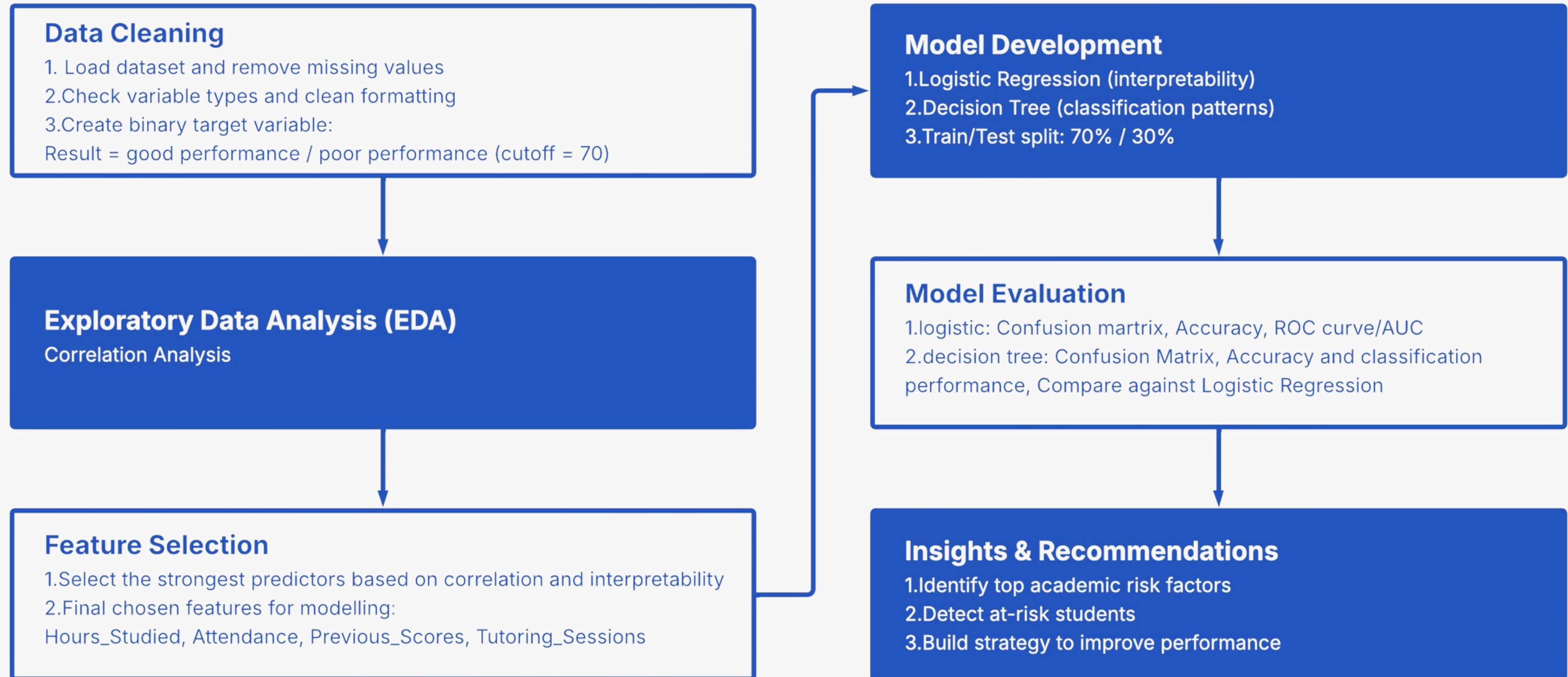


- Cleaned dataset contains **6,378 records and 19 academic + behavioural variables.**
- Created a categorical performance (result) variable for Logistic Regression.
- Set 70 as the threshold:
 - $\geq 70 \rightarrow$ Good Performance**
 - $< 70 \rightarrow$ Poor Performance**

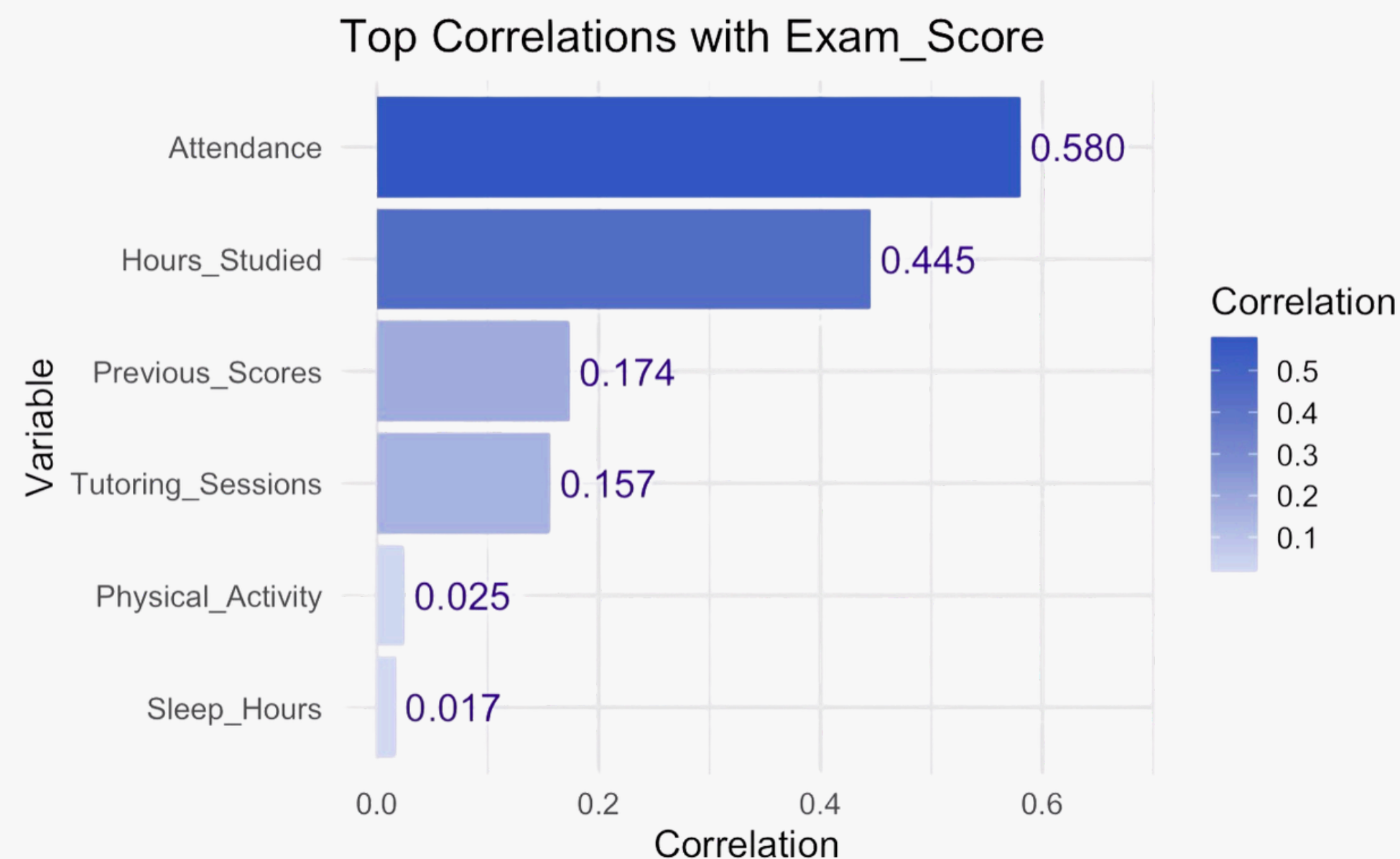


1. More Study Hours $\rightarrow$ Higher Scores
2. Higher Attendance $\rightarrow$ Better Performance
3. Through EDA, we found our strong Predictors:
 - Hours Studied
 - Attendance
 - Previous Scores
4. Models used: **Logistic Regression, Decision Tree**
5. Partitioned the data into **Train(70%) & Test sets(30%).**

Project Process



Correlation Analysis



Learning Behavior Matters Most

Attendance and study hours are the strongest predictors.

Effort > Past Ability

Current effort impacts performance more than previous scores.

Tutoring Helps, but Not Alone

Tutoring is effective only when combined with student effort.

Lifestyle Has Limited Direct Impact

Sleep and physical activity show minimal direct influence on scores.

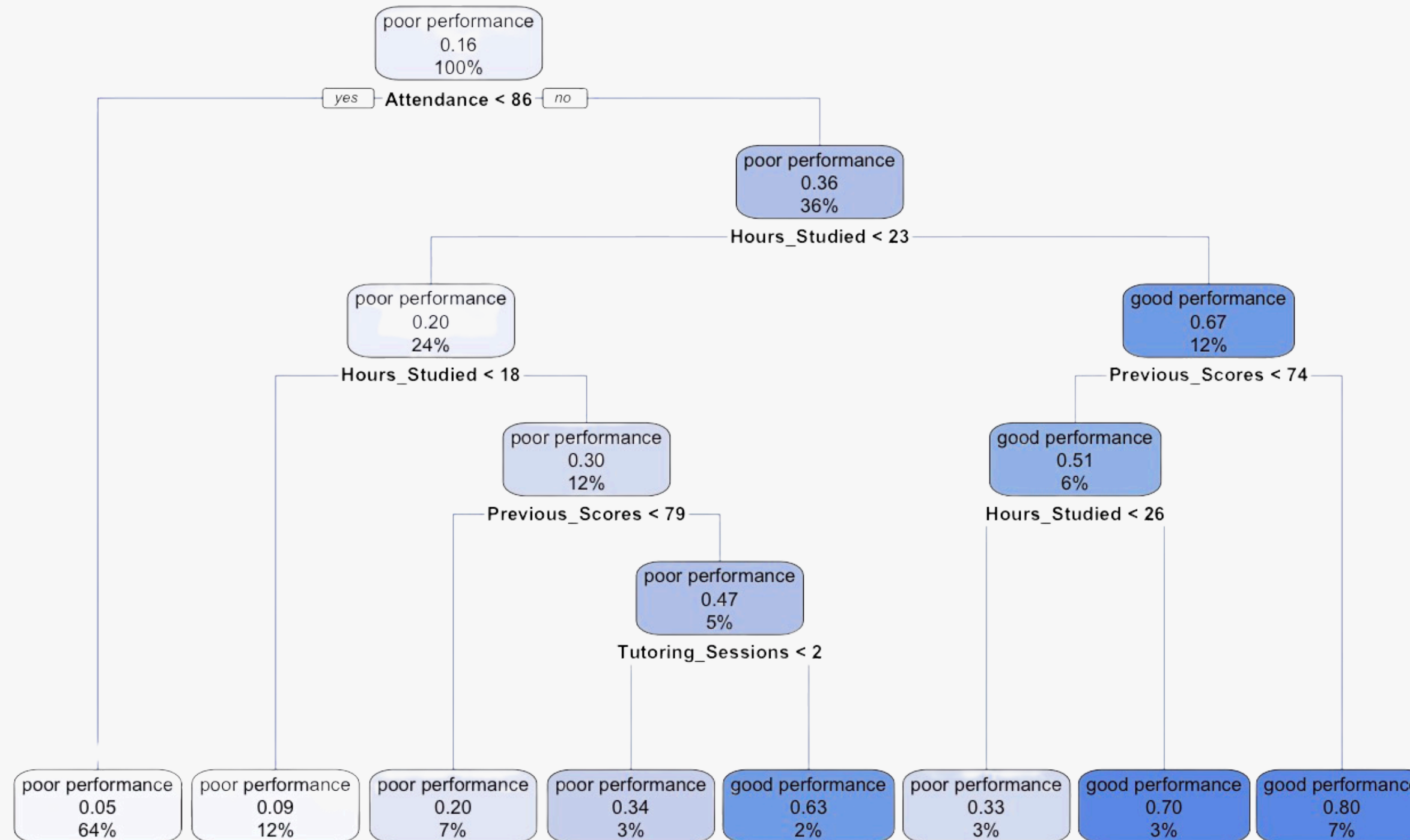
Logistic Regression

$$\text{logit}(p) = -29.665 + 0.299 * \text{Hour_studied} + 0.193 * \text{Attendance} + 0.053 * \text{Previou_scores} + 0.513 * \text{Tutorin_sessions}$$

$$p = \frac{1}{1 + e^{-(-29.665 + 0.299 * H + 0.193 * A + 0.053 * P + 0.513 * T)}}$$

	Odds Ratio	Explanation
Hours_Studied	1.348	1% increase in study hours, the odds increase by 34.8%
Attendance	1.213	1% increase in attendance, the odds raises by 21.3%
Previous_Scores	1.054	1% increase in previous exam score, the odds increases by 5.4%
Tutoring_Sessions	1.671	1% increase in tutoring session, the odds increases by 67.1%

Decision Tree Results & Driver



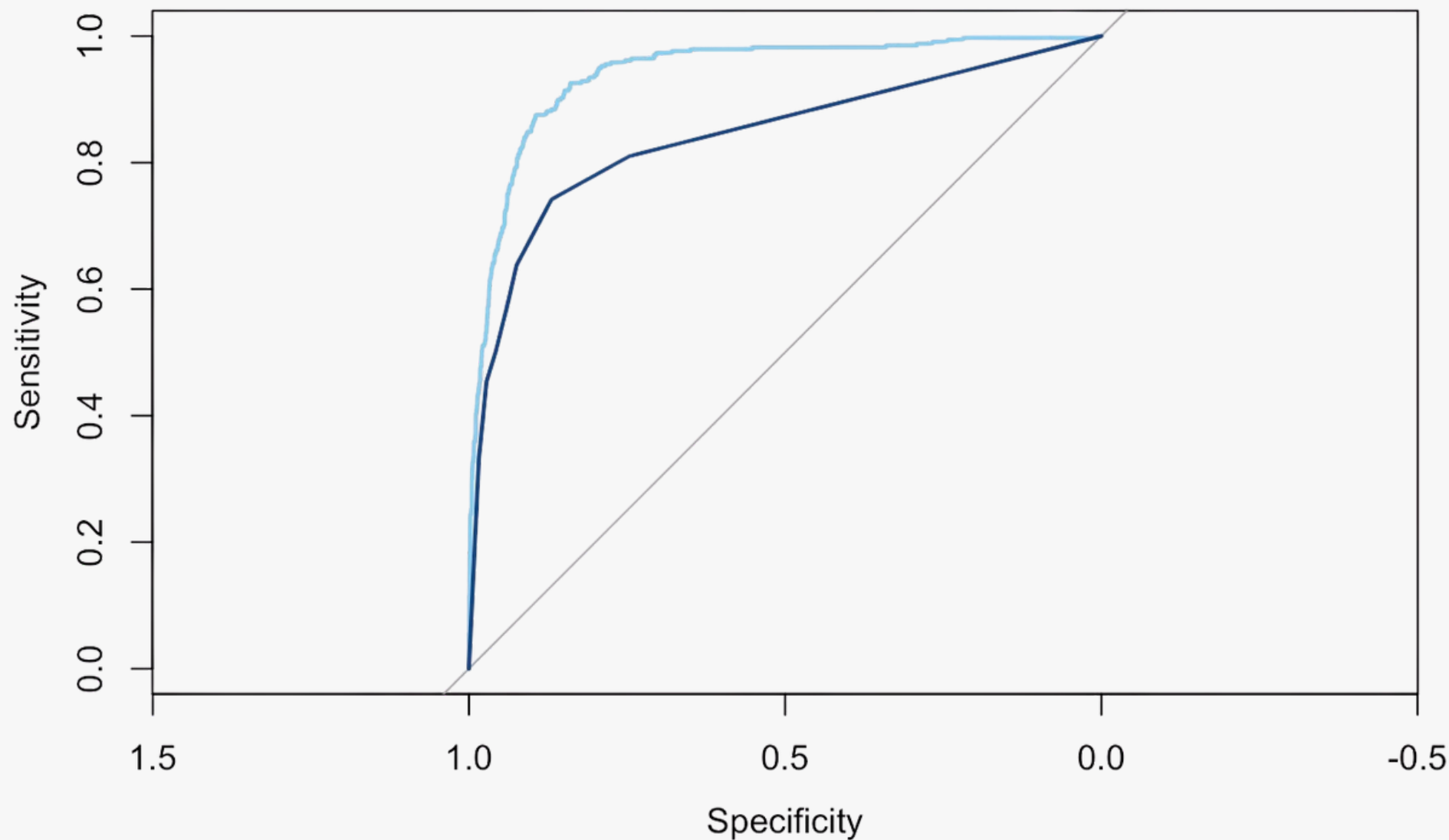
Findings:

attendance below **86%** → higher likelihood of poor performance
study hours further differentiate performance

Model Comparison

Logistic regression performs better

Metric	Logistic Regression	Decision Tree
Accuracy	90.49%	87.72%
Precision	61.42%	71.61%
Sensitivity	79.92%	50.15%
Specificity	92.15%	95.75%
AUC	Higher	Lower



Action Plan



Fix Attendance First

- Alerts for <85% attendance
- Immediate follow-up



Strengthen Study Routines

- Weekly study plans
- Study-skills workshops



Target Tutoring

- Prioritize Poor Performance students
- Link tutoring to course content



Promote Effective Behaviors

- Encourage consistent hours + participation
- Shift student mindset: effort > talent



Use Risk Levels to Allocate Support

- High-risk: close monitoring + intervention
- Medium-risk: tutoring + study planning
- Low-risk: maintain habits

**Thank You
So Much!**

GROUP 20